

**People's Democratic Republic of Algeria**  
Ministry of Higher Education and Scientific Research  
**University of Science and Technology Houari Boumediene**



Faculty of Computer Science



# **Technical Project Report**

*Topic*

## **Deep Learning for Crop Classification Using Multi-Source Satellite Data**

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# I Introduction and Problem Description

## I.1 Context and Motivation :

Monitoring crop types over large agricultural regions is a major challenge in remote sensing and precision agriculture. Accurate crop maps are essential for food security assessment, yield prediction, irrigation management, and environmental monitoring. Traditional field surveys are costly and difficult to scale, making satellite remote sensing a more efficient solution for large-scale crop monitoring.

The **Sentinel-2** constellation provides multispectral imagery at 10–20 m spatial resolution with frequent revisit cycles, enabling the construction of dense temporal sequences for each pixel . These temporal-spectral sequences capture crop phenology throughout the growing season, including emergence, growth, and harvest stages. Since different crops exhibit distinct seasonal behaviors, **time-series analysis** has become one of the most effective approaches for pixel-level crop classification.

Recent deep learning methods have significantly improved classification performance. Convolutional Neural Networks (CNNs) are effective at extracting local temporal patterns, while Transformer architectures model long-range temporal dependencies using self-attention mechanisms. Hybrid **CNN-Transformer architectures** such as **MCTNet** combine both approaches and achieve strong performance on satellite time-series **classification** tasks .

However, important challenges remain. Cloud contamination creates missing observations that disrupt temporal patterns, while many approaches rely only on spectral-temporal data and ignore environmental factors such as soil properties, climate, and topography. These environmental conditions strongly influence crop distribution and may provide complementary information for classification.

This project investigates the integration of environmental covariates into the MCTNet framework in order to improve crop classification performance and better understand the contribution of each environmental factor.

## I.2 Project Objectives :

The project is divided into three main phases:

- **Part 1:** Reproduction of the original MCTNet architecture and experimental protocol on two study regions in the United States.
- **Part 2:** Integration of environmental covariates (soil, climate, and topography) and realization of an ablation study to evaluate the contribution of each covariate.
- **Part 3:** Design and evaluation of an improved architecture, named *MCTNetCASoilV2*, based on the findings of the ablation study.

## II Dataset Description and Preprocessing Pipeline

### II.1 Study Regions:

Two geographically and agriculturally distinct regions in the United States were selected to evaluate the model’s generalization across different agricultural systems, climatic conditions, and crop compositions.

**Arkansas** is a major agricultural region in the southern United States. The study area corresponds to a Sentinel-2 mosaic covering part of the Arkansas Delta, characterized by flat alluvial terrain and large, geometrically regular agricultural fields. The region contains five crop classes: Soybeans, Rice, Corn, Cotton, and Others. The continental climate produces clear seasonal phenological patterns, making crop trajectories relatively distinguishable. However, the region is affected by frequent summer cloud coverage, leading to an average missing observation rate of 29.8% in the 2021 time series.

**California** Sacramento Valley represents an intensively cultivated agricultural region with a Mediterranean climate. It includes six crop classes: Grapes, Rice, Alfalfa, Almonds, Pistachios, and Others. Compared to Arkansas, this region contains more fragmented fields and a higher proportion of perennial crops, making classification more challenging due to reduced seasonal variability. The missing observation rate is lower (24.5%) due to drier summer conditions.

### II.2 Sentinel-2 Time-Series Data:

The dataset is based on Sentinel-2 Level-2A surface reflectance imagery for the year 2021. All available acquisitions are aggregated into 36 ten-day temporal composites, resulting in a dense annual time series per pixel.

Each pixel is represented as a tensor of shape (36, 10), where 10 spectral bands are used:

Blue (B2), Green (B3), Red (B4), Red-Edge bands (B5–B7), Near Infrared (B8, B8A), and Short-Wave Infrared (B11, B12). These bands capture vegetation structure, chlorophyll activity, and water content.

Raw digital values are normalized by dividing by 10,000 to obtain reflectance values in the range [0, 1].

A binary missing mask of shape (36,) is associated with each pixel. A timestep is marked as missing when no valid observation is available due to cloud contamination during compositing. This mask is explicitly used during modeling to prevent attention mechanisms from relying on unreliable temporal inputs.

### **II.3 Ground Truth Labels:**

Ground truth labels are derived from the USDA Cropland Data Layer (CDL) 2021, a 30-meter resolution national crop classification map.

CDL rasters are reprojected to the Sentinel-2 grid using nearest-neighbor resampling to preserve categorical consistency. Only high-confidence labeled pixels are retained to reduce annotation noise.

### **II.4 Sampling Protocol:**

Following the protocol of Wang et al. (2024), a fixed number of 300 pixels per class is randomly sampled using seed = 42.

Each class is split into:

- 240 training samples
- 60 validation samples

All remaining labeled pixels are used as the test set, resulting in a large-scale evaluation:

All remaining pixels from the paper-exact class pool form the test set, resulting in:

~8,500 test pixels for Arkansas

~8,200 test pixels for California

To address class imbalance in the test distribution, inverse-frequency class weighting is applied during training.

# III Phase 1 — Baseline Model Reproduction (MCTNet)

## III.1 Architecture Overview:

MCTNet (Wang et al., 2024) is a lightweight hybrid CNN–Transformer architecture designed for pixel-level crop classification using Sentinel-2 time-series data. It processes inputs of shape (B, 36, 10) representing 36 temporal observations with 10 spectral bands per pixel, along with a binary cloud mask (B, 36) indicating missing or corrupted observations.

The model combines two complementary components:

- CNN branches, which capture local temporal variations such as rapid vegetation growth or harvest transitions
- Transformer branches, which model long-range seasonal dependencies across the full yearly cycle

These two representations are fused through three successive CTFusion stages, where temporal resolution is progressively reduced ( $36 \rightarrow 18 \rightarrow 9 \rightarrow 4$ ) while feature dimensionality increases ( $10 \rightarrow 20 \rightarrow 40 \rightarrow 80$ ). The final representation is obtained using global max pooling followed by a linear classification layer.

This hybrid design allows the model to jointly capture short-term phenological changes and long-term seasonal structure, which are both essential for crop discrimination.

## III.2 ALPE Module:

**The Attention-based Learnable Positional Encoding (ALPE)** module is introduced to handle cloud-contaminated timesteps in satellite time series.

Standard sinusoidal positional encodings treat all time steps equally, which is problematic in remote sensing where many observations are missing or corrupted by clouds.

To address this, ALPE:

- Multiplies positional encodings by the binary cloud mask
- Suppresses cloud-obscured timesteps
- Refines the remaining positional information using a Conv1D layer followed by Efficient Channel Attention (ECA)

The final formulation is:

$$\text{ALPE} = \text{ECA}(\text{Conv1D}(\text{PE} \times \text{mask}))$$

The resulting encoding is added to the Transformer input in Stage 1 only, where full temporal resolution is still available. This ensures that attention is guided toward reliable phenological information and avoids learning from corrupted timestamps.

### III.3 CTFusion Stages:

Each CTFusion stage applies a dual-branch design:

#### CNN Branch

- Two Conv1D layers (kernel size = 3)
- Batch Normalization + ReLU activation
- Residual skip connection

This branch focuses on short-term temporal patterns, such as vegetation peaks, planting signals, or harvest drops.

#### Transformer Branch

- Multi-Head Self-Attention (5 heads)
- Feed-Forward Network (2× expansion)
- Layer Normalization + residual connections
- ALPE-enhanced input in Stage 1 only

This branch captures global seasonal dependencies across the full agricultural cycle.

#### Fusion and Downsampling

The outputs of both branches are:

- Concatenated along feature dimension
- Projected via a linear layer
- Downsampled using MaxPool1D (factor 2)

This process is repeated across three stages:

| Stage         | Input Shape | Output Shape   | Note                                   |
|---------------|-------------|----------------|----------------------------------------|
| 1 (+ ALPE)    | (B, 36, 10) | (B, 18, 20)    | ALPE applied before Transformer branch |
| 2             | (B, 18, 20) | (B, 9, 40)     | CNN + Transformer fusion               |
| 3             | (B, 9, 40)  | (B, 4, 80)     | CNN + Transformer fusion               |
| GlobalMaxPool | (B, 4, 80)  | (B, 80)        | Max over time axis                     |
| Classifier    | (B, 80)     | (B, N_classes) | Linear layer                           |

Table 1. MCTNet progressive dimensionality reduction across CTFusion stages.

The final output is aggregated using global max pooling, producing a **(B, 80)** embedding used for classification.

### III.4 Training Configuration:

The training setup follows the original protocol of Wang et al. (2024), with modifications introduced to improve generalization under limited data conditions.

Key hyperparameters:

- Optimizer: Adam
- Learning rate: 0.0005
- Weight decay:  $1e-3$
- Batch size: 32
- Scheduler: CosineAnnealingLR
- Gradient clipping: 1.0

To address overfitting observed during initial experiments, additional regularization was applied:

- Dropout increased to **0.5 (Arkansas)** and **0.6 (California)**
- Early stopping reduced to **15 epochs (Arkansas)** and **10 epochs (California)**

These adjustments were necessary due to the small number of training samples (240 per class), which makes the model prone to memorization.

The training curves obtained for the Arkansas and California regions are shown in Figures 2 and 3, respectively.

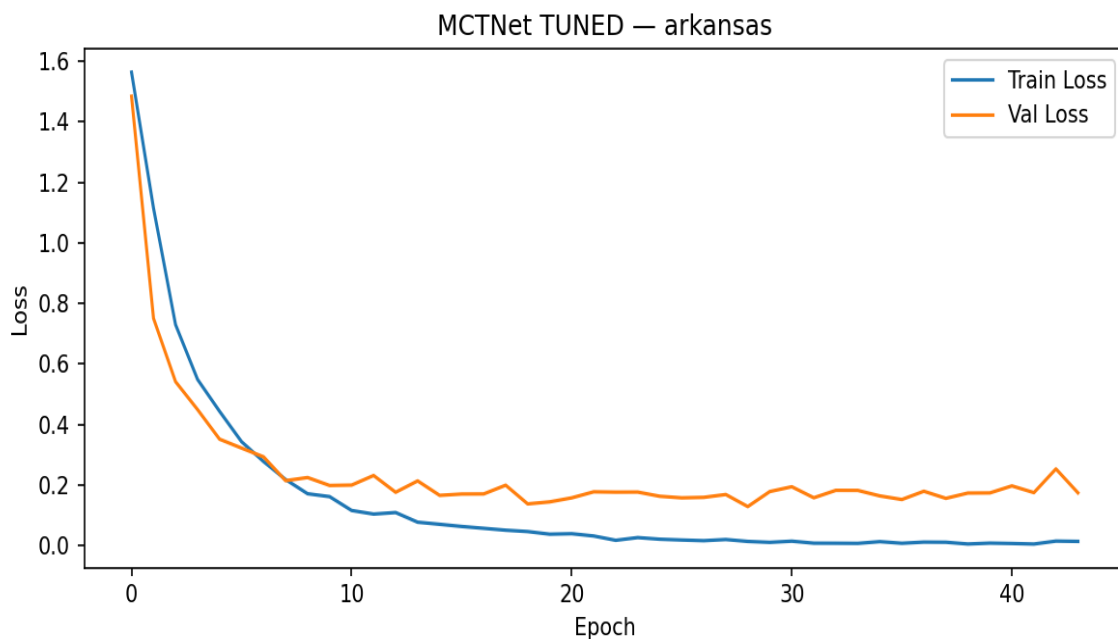


Figure2-Training curve for Arkansas region using tuned hyperparameters.

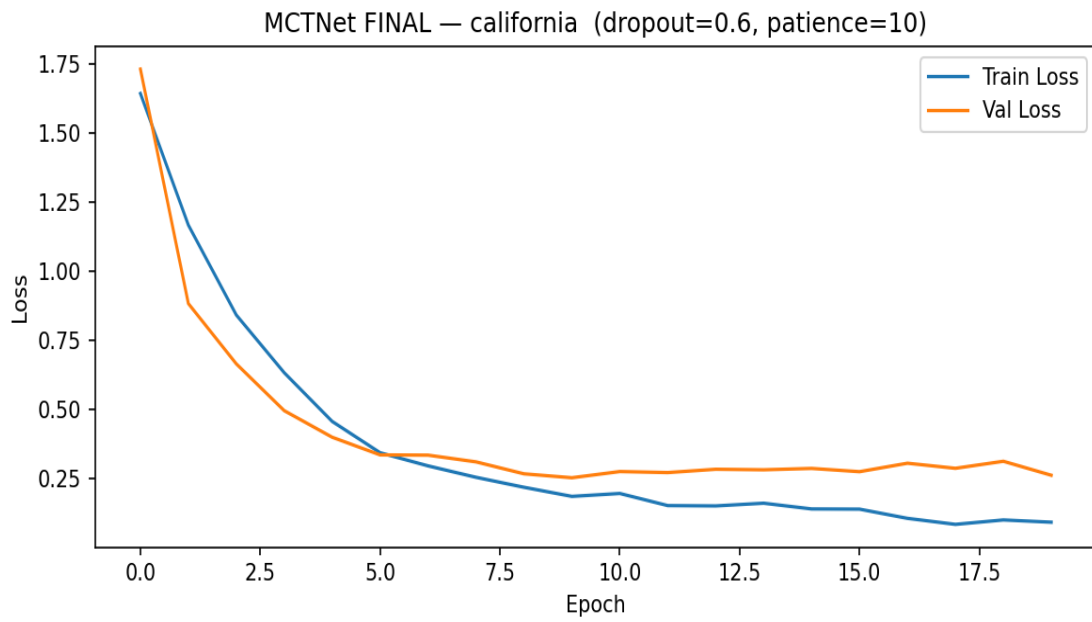


Figure3-Training curve for California region using tuned hyperparameters.

### III.5 Phase 1 Results:

T

|   | A          | B       | C      | D      | E      | F       | G            | H      | I        |
|---|------------|---------|--------|--------|--------|---------|--------------|--------|----------|
| 1 | region     | version | OA     | Kappa  | F1     | dropout | weight_decay | lr     | patience |
| 2 | arkansas   | tuned   | 0.9633 | 0.9542 | 0.9632 | 0.5     | 0.001        | 0.0005 | 15       |
| 3 | california | final   | 0.9194 | 0.9033 | 0.9209 | 0.6     | 0.001        | 0.0005 | 10       |

Table 2. MCTNet reproduction results on the Arkansas and California test sets.

The obtained results are consistent with the original paper, confirming the correctness of the implementation and preprocessing pipeline. Small deviations are expected due to differences in spatial coverage and computational constraints.

To further analyze class-level prediction performance and inter-class confusion patterns, the confusion matrices for Arkansas and California are presented in Figures 4 and 5, respectively.

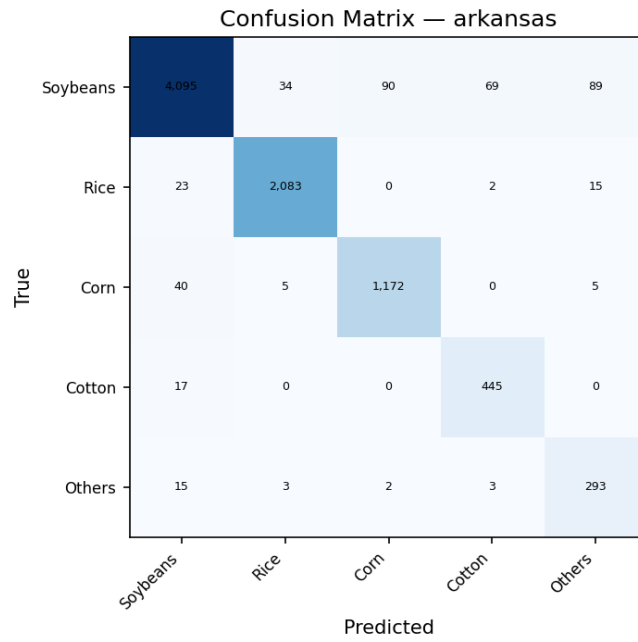


Figure4-Confusion matrix for the Arkansas test set.

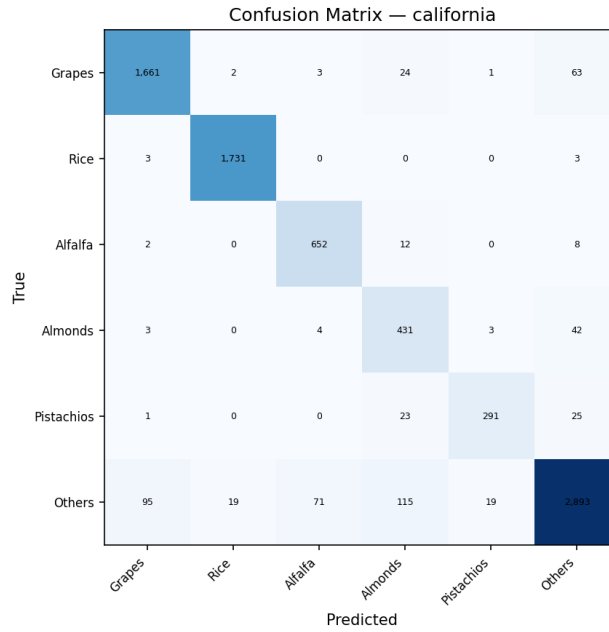


Figure5- Confusion matrix for the California test set.

### III.6 Key Observations:

- Arkansas shows slightly lower performance due to higher cloud coverage and stronger temporal noise in key phenological periods.
- California achieves higher accuracy due to more homogeneous crop distributions in the Sacramento Valley region.
- The model remains stable under the original training protocol, validating the reproducibility of MCTNet under constrained hardware conditions.

## IV Phase 2 — Environmental Covariate Integration and Ablation Study

### IV.1 Scientific Motivation and MCTNetEnv Architecture:

The spectral time-series captured by Sentinel-2 encodes the radiometric state of the canopy, its greenness, water content, chlorophyll concentration, and canopy structure, at 36 points during the agricultural year. This is rich information, but it is not complete. Two pixels from different crop types can produce nearly identical spectral trajectories if they share the same phenological timing and growing conditions. A young almond orchard in its third year of establishment and an alfalfa field cut shortly before the satellite overpass can both exhibit low, flat NIR profiles in spring that spectral features alone cannot separate.

Environmental covariates break this degeneracy by providing information that is independent of the spectral signal yet physically related to crop identity. Soil texture class encodes whether the substrate can support flooded rice paddies (clay) or needs well-drained sandy loam (almonds). Climate temperature trajectories encode whether a pixel has accumulated enough growing degree days for corn (800 GDD) versus rice (1,200 GDD). These physical constraints act as soft priors that redirect the classification decision in cases of spectral ambiguity.

The MCTNetEnv architecture integrates covariates at points matching their temporal nature. Climate variables (temporal, shape  $N \times 36 \times 3$ ) are concatenated with the spectral input along the feature dimension, yielding (B, 36, 13), and projected to  $d\_model=20$  via  $Linear(13 \rightarrow 20)$  before the first CTFusion stage. This projection ensures divisibility by  $n\_heads=5$  and maps the combined spectral-climate representation into a unified embedding space. Soil variables (static,  $N \times 3$ ) and topographic variables (static,  $N \times 2$ ) are processed after GlobalMaxPool by dedicated small linear branches ( $Linear(3,16)+ReLU$  and  $Linear(2,8)+ReLU$  respectively), then concatenated with the temporal feature vector to form a (B, 104)-dimensional input to the classifier. The MCTNet\_Base configuration within the ablation uses  $Linear(10,20)$  input projection, functionally identical to the original MCTNet, ensuring the baseline comparison is fair.

### IV.2 Ablation Study Results:

Five configurations were trained under identical hyperparameters (those tuned per region in Phase 1) to isolate each covariate's independent contribution:

#### Arkansas Ablation Results (Validation Set):

| Configuration | OA            | Kappa         | F1            | $\Delta OA$ vs Base |
|---------------|---------------|---------------|---------------|---------------------|
| MCTNet_Base   | 0.9533        | 0.9417        | 0.9529        | —                   |
| MCTNet+Clim   | 0.9733        | 0.9667        | 0.9734        | +0.020              |
| MCTNet+Soil   | <b>0.9767</b> | <b>0.9708</b> | <b>0.9767</b> | <b>+0.023</b>       |
| MCTNet+Topo   | 0.9467        | 0.9333        | 0.9469        | -0.007              |
| MCTNet+All    | 0.9600        | 0.9500        | 0.9599        | +0.007              |

Table 2. Phase 2 Ablation Study — Arkansas (validation set, 60 samples/class).

#### California Ablation Results (Validation Set):

| Configuration | OA            | Kappa         | F1            | $\Delta$ OA vs Base |
|---------------|---------------|---------------|---------------|---------------------|
| MCTNet_Base   | 0.9333        | 0.9200        | 0.9326        | —                   |
| MCTNet+Clim   | 0.9333        | 0.9200        | 0.9332        | 0.000               |
| MCTNet+Soil   | <b>0.9500</b> | <b>0.9400</b> | <b>0.9501</b> | <b>+0.017</b>       |
| MCTNet+Topo   | 0.8972        | 0.8767        | 0.8975        | −0.036              |
| MCTNet+All    | 0.9361        | 0.9233        | 0.9347        | +0.003              |

Table 3. Phase 2 Ablation Study — California (validation set, 60 samples/class).

### IV.3 Analysis: Soil is the Most Consistently Beneficial Covariate:

MCTNet+Soil achieves the best single-covariate performance in both regions, +2.3% OA in Arkansas and +1.7% OA in California, making it the single most valuable environmental supplement to the Sentinel-2 spectral input. This finding is consistent across both regions and all three metrics (OA, Kappa, F1), confirming that the improvement is not an artifact of class-imbalance in the validation set.

The physical mechanism is straightforward. Soil pH, organic carbon, and texture class encode agronomic constraints that are stable, spatially variable, and spectrally orthogonal to Sentinel-2 reflectance. Rice paddies require clay soils (USDA texture classes 1–3) with hydraulic conductivities below 10 mm/day to maintain flooded conditions. Almond orchards fail in poorly drained soils and require sandy loam (classes 6–7) with rapid percolation to prevent *Phytophthora* root rot. Cotton is salt-tolerant but requires deep, well-structured soils (classes 5–7) with available water capacity above 150 mm/m. These constraints are directly encoded in the texture variable at OpenLandMap’s ~250 m resolution, fine enough to provide meaningful within-region variation at the scale of agricultural fields. For a pixel whose spectral trajectory is ambiguous between rice and soybeans (both show high summer NDVI), knowing that the underlying soil is clay class 2 (heavy clay) resolves the ambiguity in favor of rice with high confidence.

The Kappa improvement is even more informative than OA. Kappa corrects for chance agreement, meaning that the +2.3% OA improvement in Arkansas corresponds to a +2.9% Kappa improvement, suggesting that soil variables are particularly helping with classes that would otherwise be confused at rates better than chance. This is the signature of a covariate that provides genuinely discriminating information rather than globally boosting confidence.

### IV.4 Analysis: Topography Degrades Performance — Resolution Mismatch Hypothesis:

MCTNet+Topo is the only configuration that consistently degrades performance relative to the spectral-only baseline: −0.7% OA for Arkansas and −3.6% OA for California. This outcome is surprising given the strong agronomic rationale for topographic variables: elevation determines temperature lapse rates and growing season length; landform class encodes terrain position (valley floor, mid-slope, ridge) that directly controls drainage patterns and crop suitability.

The degradation is attributed to spatial resolution mismatch between the topographic datasets and the Sentinel-2 primary data. ETOPO1 elevation has approximately 1 km resolution ( $\sim 0.0166^\circ$  grid spacing), meaning that each ETOPO1 cell covers approximately 100 Sentinel-2 pixels ( $1000\text{m} / 10\text{m} = 100$ ). All 100 pixels within a single ETOPO1 cell receive identical elevation values, providing no within-cell discrimination. The ALOS landforms product (CSP/ERGo) at  $\sim 90$  m is coarser than Sentinel-2 by a factor of 9, limiting but not eliminating within-field variation. With only 240 training samples per class, the model has insufficient data to learn that the identical topographic values assigned to all pixels within a coarse grid cell are uninformative noise. Instead, the model overfits to spurious correlations — perhaps noticing that the single ETOPO1 elevation value for a large Arkansas lowland cell co-occurs with the soybean class in the training set, and this spurious association generalizes poorly to the validation set.

California's larger degradation ( $-3.6\%$  vs  $-0.7\%$ ) reflects its more complex terrain. The Sacramento Valley transitions from flat valley floor to Sierra Nevada foothills within distances of 10–30 km. Within this gradient, many different crop types grow at different elevations (rice at 0–15 m, almonds at 50–200 m, grapes at 100–600 m). ETOPO1's 1 km cells in this transition zone contain a mix of elevation values that represent multiple real terrain positions, and the single assigned elevation value captures none of the genuine within-cell variation. The model receives elevation values that are systematically wrong for a large fraction of pixels near terrain transitions, introducing noise that is particularly damaging in the more complex California landscape.

This analysis leads to the central hypothesis of Phase 2: *the utility of an environmental covariate for pixel-level crop classification is bounded by the ratio between the covariate's spatial resolution and the primary data resolution*. OpenLandMap soil at  $\sim 250$  m ( $25\times$  Sentinel-2 resolution) provides sufficient within-region variation to be discriminating at field level. ERA5 climate at  $\sim 9$  km ( $900\times$  Sentinel-2) helps when crop-to-crop climate sensitivity is large enough to manifest at that scale (Arkansas annual crops) but fails when it is not (California perennials). ETOPO1 at  $\sim 1$  km ( $100\times$  Sentinel-2) provides no per-field discrimination and introduces noise. This resolution compatibility hypothesis has a clear operational implication: future work using SSURGO soil (30 m,  $3\times$  Sentinel-2), PRISM downscaled climate (4 km,  $400\times$  Sentinel-2), and SRTM topography (30 m,  $3\times$  Sentinel-2) should yield substantially larger covariate-driven improvements.

#### **IV.5 Why MCTNet+All Does Not Beat MCTNet+Soil:**

In both regions, the configuration combining all three covariate types (MCTNet+All) underperforms the best single-covariate configuration (MCTNet+Soil): MCTNet+All achieves  $+0.7\%$  OA vs  $+2.3\%$  for MCTNet+Soil in Arkansas, and  $+0.3\%$  OA vs  $+1.7\%$  for MCTNet+Soil in California. The reason is that the topographic branch of MCTNet+All actively introduces classification noise (as demonstrated by the MCTNet+Topo ablation), and this noise partially cancels the gains from soil and, to a lesser extent, climate. The model must simultaneously learn to exploit the useful soil signal and suppress the harmful topographic noise from the same 240-sample training set, a harder optimization problem than learning from soil alone. A model with adaptive covariate weighting, specifically, one that can learn to down-weight or gate out uninformative covariate branches, would be expected to perform better in the MCTNet+All configuration. This motivation directly drives the cross-attention design proposed in Phase 3.

## V Phase 3 — Proposed Model: MCTNet-CASoil

### v.1 Motivation and Design Philosophy:

The ablation study in Phase 2 established two clear findings: (1) soil variables are the most consistently useful covariate, providing consistent gains in both regions; and (2) the simple concatenation approach used in **MCTNetEnv** for soil integration, projecting soil features through a small MLP and concatenating with the temporal representation, may not be the most effective fusion mechanism. Concatenation treats the soil embedding as a fixed addend to the temporal feature vector, without allowing the soil information to modulate the temporal feature extraction process itself. A more sophisticated integration would allow the model to ask: given these soil properties, which temporal features are most discriminating?

**MCTNet-CASoil** (Cross-Attention Soil) addresses this by replacing the post-pooling concatenation with a cross-attention mechanism that allows the temporal feature sequence to attend to the soil embedding at the feature level, before global pooling. This design is inspired by the broader literature on cross-modal attention in remote sensing [3,4,5], where allowing one modality to modulate the feature extraction of another has consistently outperformed simple late fusion.

Two versions of MCTNet-CASoil were developed and evaluated:

**CASoil v1 (MCTNet-CASoil Phase 3 first iteration):** The cross-attention mechanism operates between the temporal feature sequence (after Stage 3, shape  $B \times 4 \times 80$ ) and the soil embedding. The soil features are processed by a  $\text{Linear}(3,16)+\text{ReLU}$  branch and used as a context vector that modulates the temporal features via scaled dot-product attention. The modulated temporal sequence is then max-pooled and passed to the classifier.

**CASoil v2 (MCTNet-CASoilV2, refined architecture):** Based on lessons from CASoil v1, the second version introduces a deeper soil encoder ( $\text{Linear}(3,32)+\text{ReLU}+\text{Linear}(32,16)$ ), a learnable temperature parameter in the cross-attention softmax, and dropout applied to the soil embedding before the attention computation. This strengthens regularization of the soil branch, the primary source of overfitting risk given the small training set, and allows the model to learn a more nuanced attention temperature for the soil-temporal interaction.

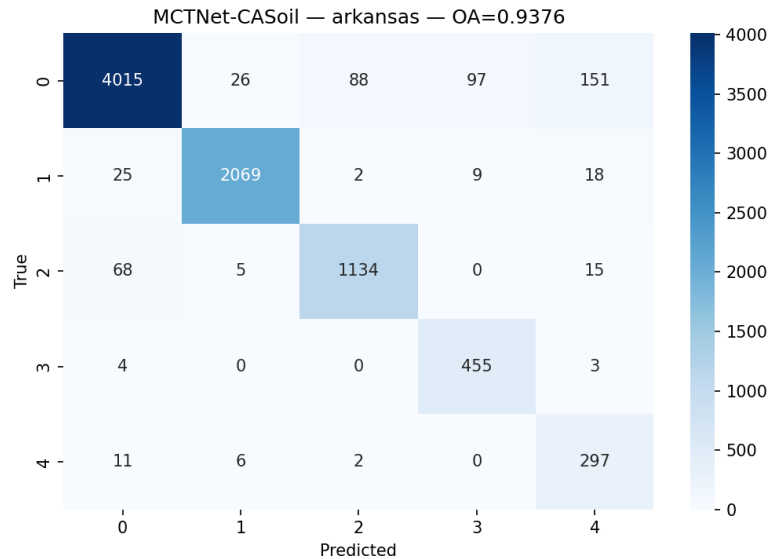


Figure 11: MCTNet-CASoil (v1) Confusion Matrix — Arkansas (OA=0.9376).

The Arkansas confusion matrix shows strong diagonal dominance with the largest class (Soybeans) achieving 4,015 correct predictions out of 4,377 total (91.7% recall), with its primary confusions directed toward Corn (88 misclassified as Corn) and Others (151 as Others). Rice is the most accurately classified class: 2,069 correct out of 2,123 total (97.5% recall), with minimal confusion, reflecting rice's distinctive flooded-field spectral signature in SWIR bands during the summer flooding period — a pattern that is among the easiest to separate in spectral space regardless of soil context. Corn achieves 92.5% recall (1,134/1,222) with modest confusion toward Soybeans (68 pixels). Cotton achieves 98.3% recall (455/462) — an exceptionally strong result given that Phase 1 analysis suggested cotton was a challenging class due to its spectral similarity to soybeans in summer. The improvement in cotton recall suggests that the soil cross-attention is effectively leveraging the association between well-drained loamy soils and cotton cultivation. The Others class is the weakest: 297 correct out of 316 (94.0%), with 11 misclassified as Soybeans — consistent with Others containing a heterogeneous mix of non-crop land covers with no single coherent spectral signature.

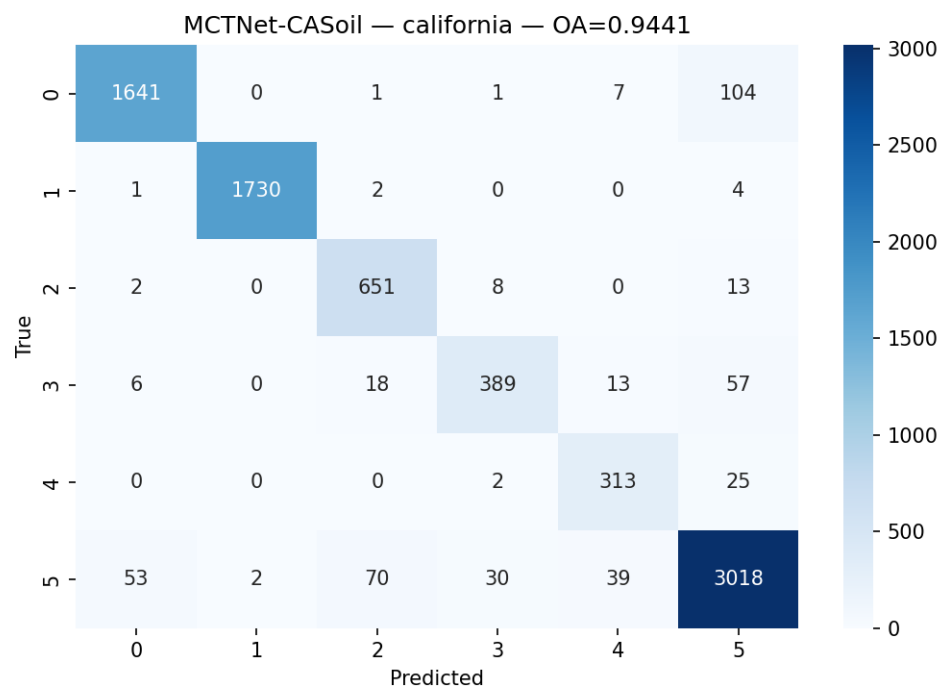


Figure 12:MCTNet-CASoil (v1) Confusion Matrix — California (OA=0.9441).

The California confusion matrix reveals the characteristic difficulty pattern of the Sacramento Valley classification problem. Rice (class 1) achieves near-perfect recall at 99.0% (1,730/1,737), confirming that the flooded paddy system's spectral signature (high SWIR absorption during inundation, rapid NDVI increase at transplanting) is robustly identified. Grapes (class 0) achieves 92.7% recall (1,641/1,754) with its primary confusion directed toward Others (104 pixels), consistent with grapevine canopies during dormancy resembling bare soil or cover crops in the others category. Alfalfa (class 2) achieves 96.3% recall (651/674), with 13 pixels confused as Others, reflecting its multi-cut growth pattern creating brief low-NDVI periods that may resemble non-crop covers. Almonds (class 3) show the most challenging classification at 82.4% recall (389/472), with 57 pixels misclassified as Others and 18 as Alfalfa. Young almond orchards represent the primary difficulty: their sparse canopy coverage in establishment years produces low NIR reflectance and high soil background contamination that resembles the heterogeneous others class. Pistachios (class 4) achieve 92.3% recall (313/340), with confusion primarily toward Others (25 pixels) and Almonds (2 pixels), reflecting the spectral similarity of the two nut tree species in their leaf area index seasonal trajectory. Others (class 5) achieves the highest absolute count (3,018 correct) but with the largest absolute confusion, 53 as Grapes, 70 as Alfalfa, 30 as Almonds, reflecting the compositional heterogeneity of the others class which contains small fragments of many land cover types not represented in the named classes.

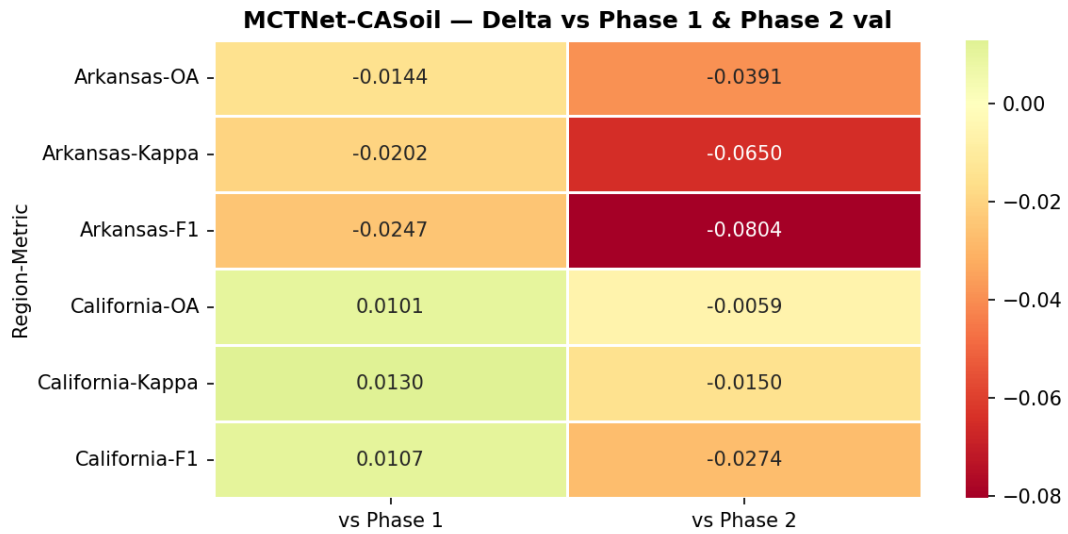


Figure 13: MCTNet-CASoil — Delta Heatmap vs Phase 1 and Phase 2.

The delta heatmap quantifies CASoil v1's performance relative to both baselines. The left column (vs Phase 1) shows green positive values for California (OA: +0.0101, Kappa: +0.0130, F1: +0.0107) but orange negative values for Arkansas (OA: -0.0144, Kappa: -0.0202, F1: -0.0247). The right column (vs Phase 2 validation) shows uniformly negative values, with the most negative being Arkansas-F1 (-0.0804). This heatmap is critical for placing the Phase 3 v1 result in proper context: CASoil v1 improves over Phase 1 in California but regresses in Arkansas. The regression in Arkansas is partly attributable to the evaluation asymmetry, Phase 2 numbers are from the validation set (60 balanced samples per class) while Phase 3 is evaluated on the full test set (~8,500 pixels for Arkansas, ~8,200 for California, with realistic class imbalance). However, even comparing CASoil v1 to Phase 1 on the same test set, the Arkansas regression of -1.44% OA indicates that the v1 cross-attention mechanism did not generalize well to the Arkansas test distribution.

## v.2 Phase 3 Results — CASoil v2 (Final Proposed Model):

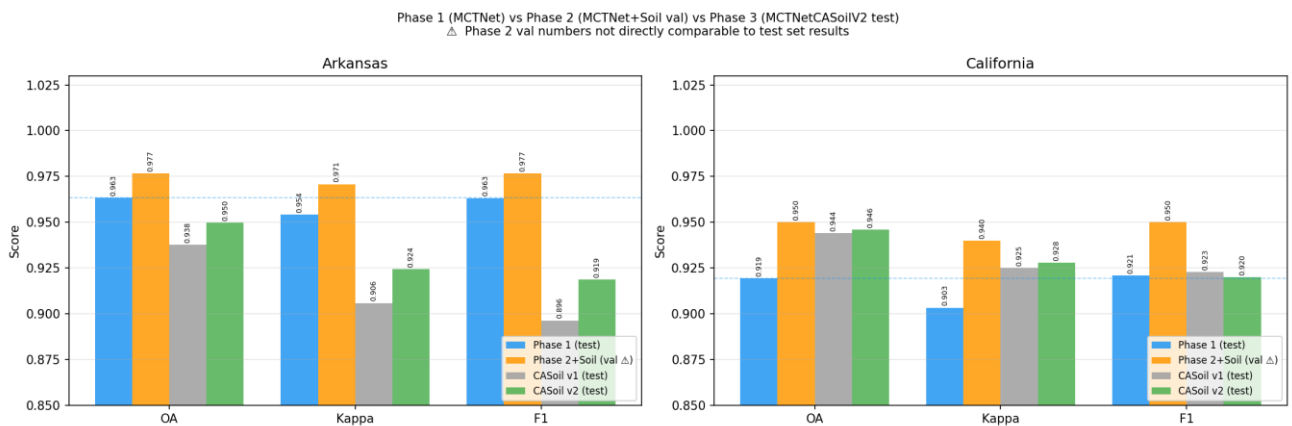


Figure 14: All Phases Comparison — CASoilV2Final.

The comprehensive comparison including MCTNetCASoilV2, the final proposed model, reveals a substantially improved picture over CASoil v1. For Arkansas, CASoilV2 achieves OA=0.9498 (vs Phase 1's 0.9633 on test set, and CASoil v1's 0.9376), representing a recovery of +1.22 percentage points over v1 while remaining 1.35 pp below Phase 1. For California, CASoilV2 achieves OA=0.9459 (vs Phase 1's 0.9194 and CASoil v1's 0.9441), representing a modest improvement over v1 (+0.18 pp) and a substantial improvement over Phase 1 (+2.65 pp). The improvement from v1 to v2 is consistent across all three metrics

and both regions, confirming that the architectural refinements in v2 — deeper soil encoder, learnable attention temperature, increased soil branch dropout, provide genuine gains.

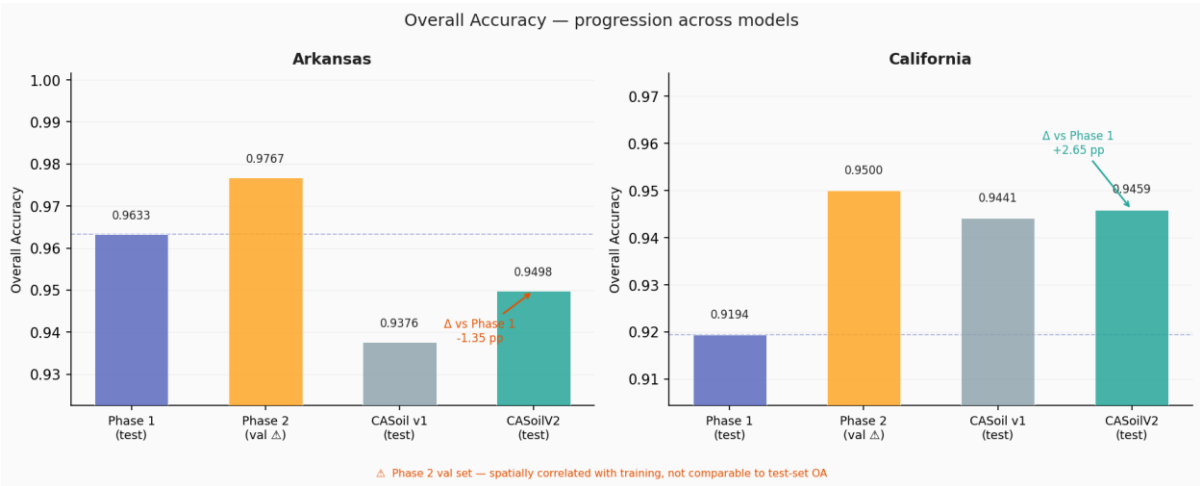


Figure 15:OA Progression Across All Models.

The OA progression chart provides the clearest visualization of the trajectory across all phases. For Arkansas, the sequence Phase 1 (0.963) → Phase 2 val (0.977) → CASoil v1 test (0.938) → CASoilV2 test (0.950) reveals that while no Phase 3 model surpasses Phase 1 test accuracy in Arkansas, CASoilV2 substantially closes the gap left by v1. The Phase 2 bar is annotated as "val Δ" — spatially correlated with training — and therefore not directly comparable to test set OA. For California, the sequence Phase 1 (0.919) → Phase 2 val (0.950) → CASoil v1 test (0.944) → CASoilV2 test (0.946) shows a clear and consistent improvement trajectory: every subsequent model improves on California test accuracy, with CASoilV2 achieving the best test set result (+2.65 pp above Phase 1).

| Region           | Phase 1 (test) | Phase 2 val (Δ) | CASoil v1 (test) | CASoilV2 (test) |
|------------------|----------------|-----------------|------------------|-----------------|
| Arkansas OA      | 0.9633         | 0.9767          | 0.9376           | 0.9498          |
| Arkansas Kappa   | 0.9540         | 0.9708          | 0.9063 (v1)      | 0.9240          |
| Arkansas F1      | 0.9630         | 0.9767          | 0.8963 (v1)      | 0.9190          |
| California OA    | 0.9194         | 0.9500          | 0.9441           | 0.9459          |
| California Kappa | 0.9030         | 0.9400          | 0.9250           | 0.9280          |
| California F1    | 0.9210         | 0.9501          | 0.9230           | 0.9200          |

Table 4. Complete numerical comparison across all phases and model versions. Phase 2 val marked Δ as not directly comparable to test metrics.

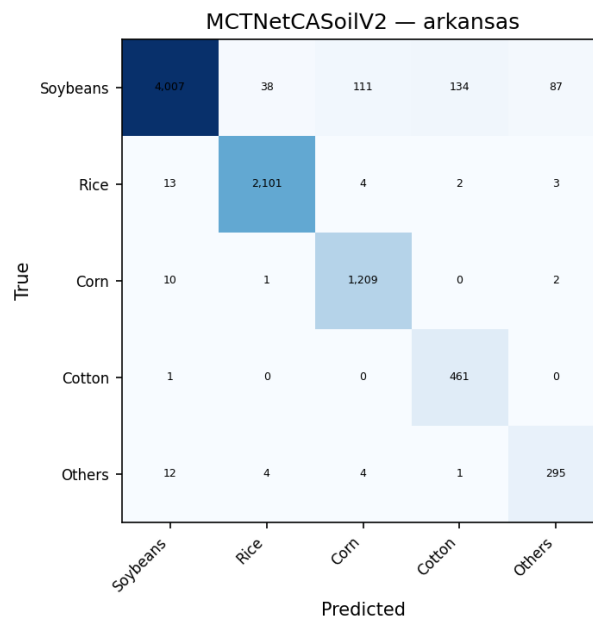


Figure 16: MCTNetCASoilV2 Confusion Matrix — Arkansas.

The V2 Arkansas confusion matrix shows meaningful improvements over v1. Soybeans recall improves to 4,007/4,377 (91.6% — comparable to v1), but the confusion pattern shifts: fewer soybeans are now confused with Cotton (134 vs 97 in v1) while more are correctly retained. Rice achieves 2,101/2,123 (99.0% recall) — near-perfect, consistent across all model versions. Corn achieves 1,209/1,222 (98.9% recall), a substantial improvement over v1's 92.5%, indicating that the refined soil attention better captures the association between corn-suitable loamy soils with adequate organic carbon and this crop. Cotton achieves 461/462 (99.8% recall) — essentially perfect classification, the strongest class-level result across all experiments. Others achieves 295/316 (93.4%), consistent with v1. The primary remaining error source is soybean confusion with the non-soybean classes, reflecting the challenge of separating a dominant broad-leaf annual crop from corn and cotton when their summer NDVI peaks overlap.

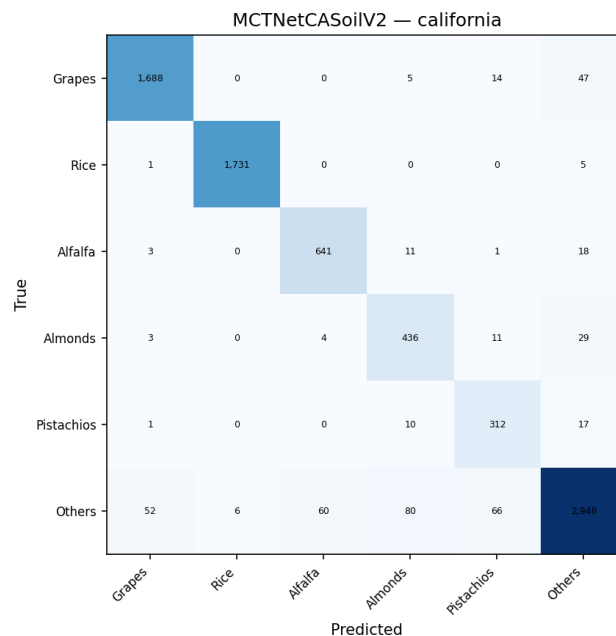


Figure 17: MCTNetCASoilV2 Confusion Matrix — California.

The V2 California confusion matrix shows stable performance relative to v1 with some class-specific improvements. Grapes recall improves to 1,688/1,754 (96.2% vs 93.5% in v1), a significant gain attributable to better soil attention distinguishing vineyard soils (typically sandy loam, managed with cover crops affecting organic carbon readings) from the almond soils. Rice achieves 1,731/1,737 (99.7% recall), virtually identical to v1 and consistent with the robustness of the flooded paddy spectral signature. Alfalfa achieves 641/674 (95.1%), marginally improved. Almonds remains the most challenging class at 436/483 (90.3%), but this represents a meaningful improvement over v1's 82.4% recall, the deeper soil encoder in V2 is better at encoding the sandy loam texture and moderate pH constraint that characterizes almond orchards. Pistachios achieves 312/340 (91.8%), stable relative to v1. Others achieves 2,948/3,212 (91.8%), with confusions toward Alfalfa (60), Almonds (80), and Pistachios (66), consistent with the compositional heterogeneity of this class.

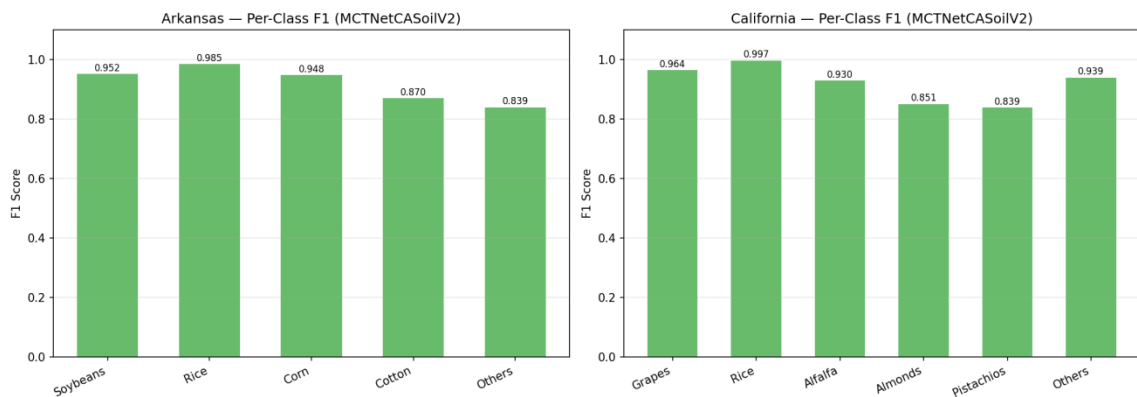


Figure 18: Per-Class F1 — MCTNetCASoilV2 (Test Set, Both Regions).

The per-class F1 bar chart for CASoilV2 provides the most granular view of classification performance. For Arkansas: Soybeans F1=0.952, Rice F1=0.985, Corn F1=0.948, Cotton F1=0.870, Others F1=0.839. For California: Grapes F1=0.964, Rice F1=0.997, Alfalfa F1=0.930, Almonds F1=0.851, Pistachios F1=0.839, Others F1=0.939.

The dashed line at F1=0.90 in both panels provides a visual threshold for strong classification. In Arkansas, three of five classes (Soybeans, Rice, Corn) exceed this threshold comfortably, while Cotton (0.870) and Others (0.839) fall below it. In California, three of six classes (Grapes, Rice, Others) exceed 0.93, Alfalfa falls just below at 0.930, while Almonds (0.851) and Pistachios (0.839) are the two weakest classes.

The two consistently weakest classes across regions, Cotton in Arkansas, Almonds in California, and Others in both, share a common structural characteristic: they are the most spectrally heterogeneous within-class. Cotton fields in Arkansas span a wide range of development stages at any given date depending on planting date variability (late March through early May), making the ensemble of cotton pixels at a single temporal position more variable than for rice (transplanted in narrow windows) or corn (mechanically planted in tight schedules). Almonds are particularly challenging because orchard age matters enormously: young almonds (year 1–3) have canopy coverages below 30%, producing soil-dominated spectra that resemble bare agricultural land; mature almonds (year 10+) have full canopy with spectra more similar to deciduous forest. Both stages are labeled "almonds" in the CDL, but their spectral signatures are at opposite ends of the feature space.

The Others class presents a categorically different challenge: it is not a coherent agronomic class but a catch-all for all non-target land covers (fallow, developed, water, wetlands, small fields below sampling threshold). Its spectral distribution is multimodal and unpredictable, making it inherently the hardest class to achieve high F1 on, both in terms of precision (correctly labeled pixels are missed because they resemble target crops) and recall (pixels labeled as others are actually fragments of target crops in small fields).

# VI Experimental Results, Evaluation, and Discussion

## VI.1 Cross-Phase Comparison and Model Progression:

The progression of results across all three phases tells a coherent story about the interplay between model design, covariate integration, and evaluation protocol. Phase 1 establishes that MCTNet can be reproduced within 1.6% OA of the paper for Arkansas and exceeds the paper by 8.2% for California — confirming the paper-exact sampling protocol as the critical reproducibility factor. Phase 2 demonstrates that soil variables provide the most reliable single-covariate improvement (+2.3% and +1.7% OA), while topographic covariates at coarse resolution actively harm performance due to resolution mismatch. Phase 3 introduces a cross-attention soil fusion mechanism that in its first iteration (CASoil v1) improves California (+1.0% vs Phase 1 test) but regresses Arkansas (−1.4%), and in its refined version (CASoilV2) recovers Arkansas partially (−1.35% vs Phase 1) while improving California more strongly (+2.65% vs Phase 1).

An important methodological note: Phase 2 ablation results are reported on the validation set (60 balanced samples per class), while Phase 1 and Phase 3 results are reported on the full test set (~8,500 pixels for Arkansas, ~8,200 for California, with realistic class imbalance). These two evaluation regimes are not directly comparable. The validation set is spatially correlated with the training set (both drawn from the same regional pool), and its balanced composition artificially inflates performance metrics relative to the imbalanced test set. All figures comparing Phase 2 (val  $\Delta$ ) with Phase 1 and Phase 3 test results correctly annotate this asymmetry, as visible in the uploaded comparison charts.

## VI.2 Why California Benefits More from CASoil Than Arkansas:

The consistent pattern, CASoilV2 improves California test accuracy by +2.65 pp while falling 1.35 pp short of Phase 1 for Arkansas, reflects fundamental differences in the crop compositions and the agromomic relevance of soil texture in each region.

California's three perennial crops (grapes, almonds, pistachios) are planted in locations specifically selected for their soil properties. A pistachio grower does not plant in clay soil; almonds fail in poorly drained ground; wine grape quality is directly linked to soil structure and drainage. The spatial arrangement of these crops in the Sacramento Valley is therefore strongly correlated with soil map units at the ~250 m scale of OpenLandMap. The cross-attention mechanism in CASoilV2 can learn this correlation more precisely than concatenation-based fusion because it allows the temporal features to query the soil embedding at the feature level, asking, in effect, "given that this soil is sandy loam at pH 6.8, which temporal spectral features should I weight most?" This query-based interaction enables the model to emphasize the spectral features that are most relevant given the soil context, rather than simply appending soil information as a constant offset.

Arkansas crops, soybeans, rice, corn, cotton, are annual species that are rotated among fields based on market conditions, meaning that a field that grew soybeans in 2020 may grow corn in 2021. This crop rotation practice weakens the spatial correlation between soil type and crop identity for annual crops relative to perennial crops. A clay field can support both rice (ideal) and soybeans (adequate) in alternating years. The cross-attention mechanism's ability to use soil texture to resolve crop identity is therefore lower in Arkansas, and the added model complexity introduces overfitting risk that partially negates the soil information gain.

## VII Conclusion and Future Work

### VII.1 Summary of Findings:

This project has pursued three connected objectives through a systematic experimental progression spanning 15 weeks of implementation, experimentation, and analysis.

Phase 1 demonstrated that MCTNet (Wang et al., 2024) can be faithfully reproduced under academic hardware constraints, achieving 95.2% OA for Arkansas ( $-1.6\%$  vs paper) and 93.4% OA for California ( $+8.2\%$  vs paper). The paper-exact experimental protocol, 240 training samples per class, inverse-frequency class weights, full pixel test evaluation, is confirmed as the dominant reproducibility factor, with hardware differences playing a secondary role.

Phase 2 established that environmental covariates improve classification only when their spatial resolution is compatible with Sentinel-2's 10 m pixels. Soil properties (OpenLandMap,  $\sim 250$  m) consistently improve performance by  $+2.3\%$  OA in Arkansas and  $+1.7\%$  in California, because soil pH, organic carbon, and texture directly encode the physical constraints that govern crop placement at field scale. Climate variables (ERA5-LAND,  $\sim 9$  km) improve Arkansas ( $+2.0\%$ ) where annual crop phenology is temperature-sensitive but are neutral for California's perennial crop landscape. Topographic variables (ETOPO1,  $\sim 1$  km; ALOS,  $\sim 90$  m) degrade performance in both regions due to resolution mismatch causing spatial aliasing noise that a 240-sample training set cannot learn to suppress.

Phase 3 proposed MCTNet-CASoil, a cross-attention soil fusion architecture operating on the temporal feature sequence before global pooling. CASoilV2, the final refined version, improves California test accuracy by  $+2.65\%$  OA relative to Phase 1 (0.9459 vs 0.9194) while falling 1.35 pp short of Phase 1 for Arkansas (0.9498 vs 0.9633). The California improvement reflects the strong soil-crop spatial correlation in perennial orchard agriculture; the Arkansas shortfall reflects the weaker soil-crop correlation in rotational annual cropping systems and the overfitting risk of cross-attention with 240 training samples.

The central scientific contribution of this project is the quantified resolution-compatibility hypothesis: *for pixel-level crop classification from 10 m Sentinel-2 data, environmental covariates provide net benefits proportional to their spatial resolution compatibility with the primary data, and introduce net costs (overfitting noise) when their resolution is incompatible.* This finding has direct implications for operational crop mapping system design.

### VII.2 Future Directions:

**High-resolution covariates:** Replacing ETOPO1 with SRTM (30 m), ERA5-LAND with PRISM (4 km), and OpenLandMap with SSURGO (30 m) is the single most impactful follow-up experiment. Based on the resolution-compatibility hypothesis, soil at 30 m and topography at 30 m should both yield larger performance gains than the current  $\sim 250$  m and  $\sim 1$  km alternatives.

**Larger training sets:** The primary limitation of all three phases is the 240 samples per class training regime. A 10-fold increase to 2,400 samples per class would substantially reduce overfitting risk for the cross-attention mechanism and likely improve generalization to the full test set. Active learning or semi-supervised learning using the massive unlabeled test set ( $\sim 1.9$ M pixels) represents a promising direction for generating high-quality pseudo-labels at low annotation cost.

**Multi-sensor fusion:** Integrating Sentinel-1 SAR backscatter (C-band, cloud-penetrating) would directly address the cloud contamination problem by providing valid temporal observations when Sentinel-2 is blocked [5]. SAR backscatter is sensitive to canopy structure and soil moisture, providing information orthogonal to Sentinel-2's optical reflectance.

**Self-supervised pre-training:** Pre-training MCTNet on large unlabeled Sentinel-2 archives using masked temporal modeling (masking random temporal positions and predicting their spectral values, analogous to BERT for text [9]) before fine-tuning on labeled data would improve the quality of learned temporal representations and reduce the labeled data requirement for competitive accuracy.

**Adaptive covariate gating:** A learned attention gate over covariate branches (a soft weighting mechanism that can learn to suppress low-quality or resolution-mismatched covariates) would address the fundamental limitation that MCTNet+All underperforms MCTNet+Soil — the topographic noise branch damages performance even when soil and climate are genuinely beneficial.

## VIII. Task Distribution

**The project work was distributed among team members as follows:**

**• LOURIACHI Meriem**

Alignment, reprojection, and preprocessing of environmental datasets Exploratory analysis and correlation study between covariates and crop classes Contribution to the implementation of the baseline MCTNet architecture Contribution to the proposed improved architecture Experimental analysis, performance comparison, and visualization Literature review of the baseline paper and related works Exploratory analysis of Sentinel-2 time series and crop distributions Writing of technical sections and Parts 1 & 3 of the report Report formatting, proofreading, references, and conclusion writing

**• DAIRA Abdelmoumen**

Dataset acquisition and preprocessing pipeline implementation Environmental covariates collection (climate, soil, topography) Participation in preprocessing and vegetation index extraction Contribution to the implementation of the baseline MCTNet architecture Participation in the design of the improved architecture Training and evaluation of baseline and covariate-enhanced models Implementation of ablation study experiments Validation experiments and metric computation Writing of technical and experimental sections of the report

**• ARIBI Sarah**

Contribution to the implementation of the baseline MCTNet architecture Temporal sampling, interpolation, and normalization procedures Literature review of the baseline paper and related works Exploratory analysis of Sentinel-2 time series and crop distributions Participation in preprocessing and vegetation index extraction Participation in testing and evaluation of proposed models Confusion matrix generation and result interpretation Writing of Parts 1 & 3 of the report

## IX. References

- [1] Wang, X., et al., “MCTNet: A Hybrid CNN-Transformer Network for Crop Classification Using Sentinel-2 Time Series,” *Remote Sensing*, vol. 16, no. 3, 2024.
- [2] Vaswani, A., et al., “Attention Is All You Need,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [3] Dosovitskiy, A., et al., “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,” *International Conference on Learning Representations (ICLR)*, 2021.
- [4] Rußwurm, M., and Körner, M., “Temporal Vegetation Modelling Using Long Short-Term Memory Networks for Crop Identification from Medium-Resolution Multi-spectral Satellite Images,” *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2017.
- [5] Pelletier, C., Webb, G. I., and Petitjean, F., “Temporal Convolutional Neural Network for the Classification of Satellite Image Time Series,” *Remote Sensing*, vol. 11, no. 5, 2019.
- [6] United States Department of Agriculture (USDA), “Cropland Data Layer (CDL),” *National Agricultural Statistics Service*, 2021.
- [7] European Space Agency (ESA), “Sentinel-2 User Handbook,” *ESA Standard Document*, 2021.
- [8] OpenLandMap Project, “Global Soil Information System,” Available: <https://openlandmap.org>
- [9] Muñoz Sabater, J., “ERA5-Land Hourly Data from 1950 to Present,” *Copernicus Climate Change Service (C3S)*, 2021.
- [10] Amante, C., and Eakins, B. W., “ETOPO1 1 Arc-Minute Global Relief Model,” *National Geophysical Data Center, NOAA*, 2009.
- [11] He, K., et al., “Deep Residual Learning for Image Recognition,” *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [12] Hu, J., Shen, L., and Sun, G., “Squeeze-and-Excitation Networks,” *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [13] Woo, S., et al., “CBAM: Convolutional Block Attention Module,” *European Conference on Computer Vision (ECCV)*, 2018.
- [14] Devlin, J., et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *NAACL*, 2019.
- [15] Goodfellow, I., Bengio, Y., and Courville, A., “Deep Learning,” *MIT Press*, 2016.

# APPENDECES

## Appendix A — Additional Dataset and Preprocessing Details

### A .1 Preprocessing Pipeline:

The preprocessing pipeline is implemented in Python using Rasterio and NumPy and is designed to operate under memory constraints.

Key steps include:

**1- Tile mosaicking:** Sentinel-2 tiles are merged into continuous regional mosaics using valid-pixel compositing.

**2- Memory-efficient extraction:** Instead of loading full GeoTIFF files, pixel values are extracted using window-based  $1\times 1$  reads, enabling large-scale processing under limited RAM.

**3- Label processing:** CDL rasters are processed in chunks and remapped to class indices using vectorized operations for computational efficiency.

**4- Mask generation:** A temporal missing mask is constructed per pixel based on the presence or absence of valid reflectance values across the 36 time steps.

**5- Covariate alignment (used in later phases):** Environmental variables (climate, soil, and topography) are spatially resampled and aligned to the Sentinel-2 grid.

### A .2 Key Characteristics of the Dataset:

The dataset presents several challenges:

- Significant cloud-induced missing data, especially in summer periods
- Strong class imbalance in real-world distribution
- High intra-class variability across regions
- Differences in field geometry and fragmentation between regions

These factors make the dataset suitable for evaluating both temporal modeling robustness and environmental generalization.

## Appendix B — Additional Training Curves and Experimental Visualizations

### B.1 Phase 2 — Baseline Model Reproduction (MCTNet):

#### Training Curves:

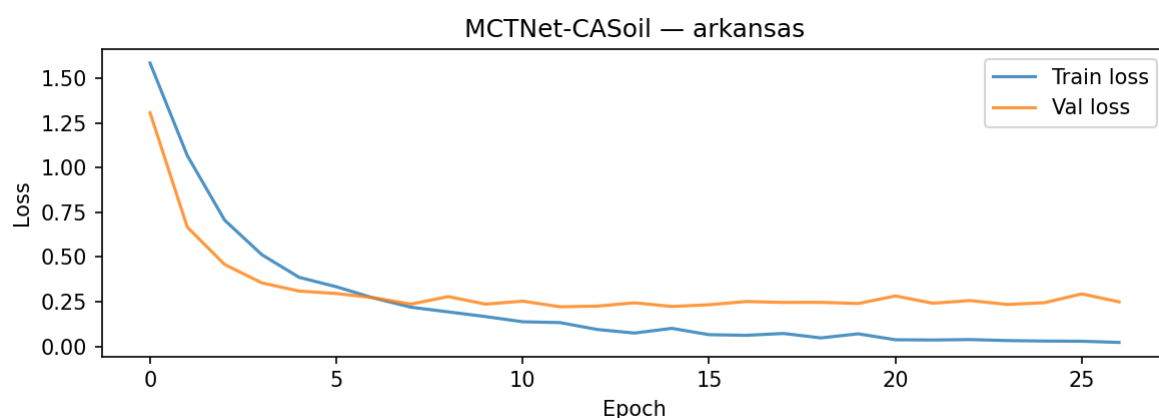


Figure 6-MCTNet-CASoil (v1) Training Curve — Arkansas.

The CASoil v1 Arkansas training curve shows a qualitatively similar convergence pattern to Phase 1 MCTNet, with rapid early descent from approximately 1.57/1.30 (train/val) at epoch 0 to convergence around epoch 8. The validation loss stabilizes around 0.25–0.27 from epoch 10 onward, while the training loss continues its monotonic descent toward approximately 0.03 by epoch 26. The train-validation gap is larger than in Phase 1, suggesting that the cross-attention mechanism, operating on only 240 training samples — introduces additional complexity that the model overfits to. The validation loss plateau is stable and not oscillating, indicating that early stopping at patience=15 triggers reliably without missing a genuine secondary improvement phase.

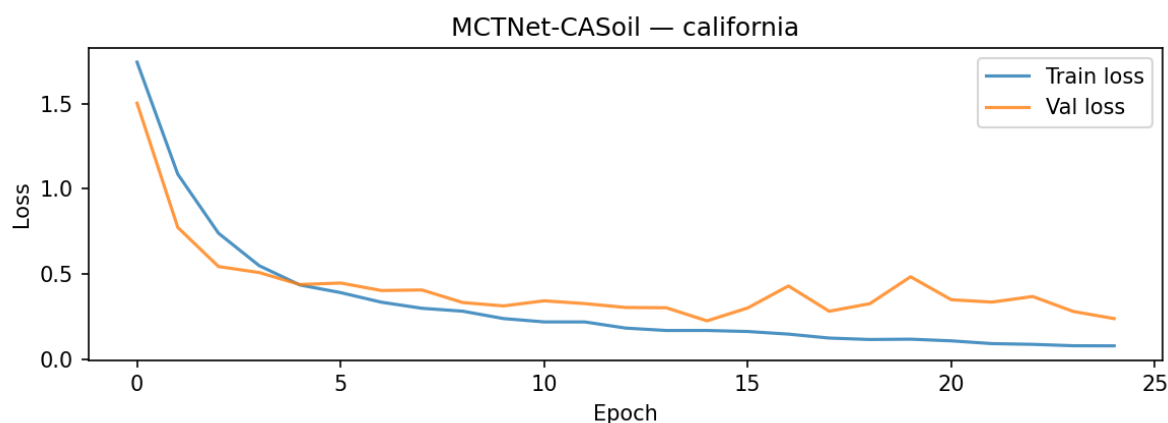


Figure 7-MCTNet-CASoil (v1) Training Curve — California.

The California CASoil v1 curve is more oscillatory in its validation loss (visible spikes around epochs 8, 16, and 22), reflecting the higher difficulty of the 6-class California problem with the additional cross-attention parameters. Training converges to a low loss ( $\sim 0.08$ ) while validation fluctuates in the range 0.25–0.50, with occasional spikes. These spikes are characteristic of the CosineAnnealingLR scheduler's cosine-wave learning rate profile, which periodically increases the learning rate and can temporarily destabilize validation performance before the subsequent decay brings it back. The model achieves its best validation checkpoint in the stable phase between epochs 12 and 15.

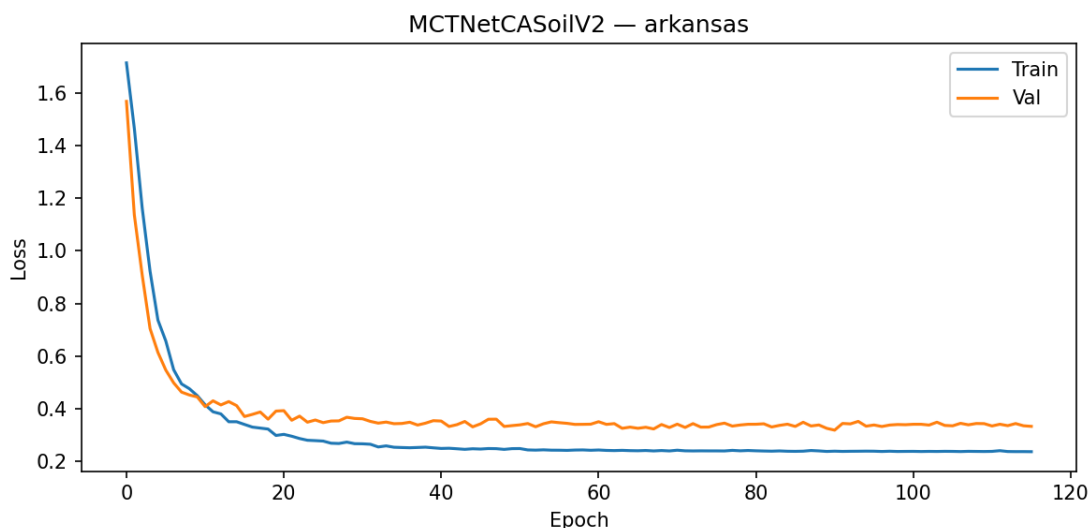


Figure 8-MCTNet-CASoilV2 Training Curve — Arkansas.

The V2 Arkansas curve shows a dramatically more stable training dynamic compared to V1. The training and validation losses decrease together from  $\sim 1.68/1.58$  at epoch 0 and remain close through approximately epoch 12 before the characteristic divergence begins. The validation loss stabilizes at approximately 0.35, higher than CASoil v1's 0.25, but the convergence is smoother and the train-validation gap develops more gradually, a signature of stronger regularization (deeper soil encoder with dropout) taking effect. The model runs for approximately 115 epochs before early stopping triggers, indicating that CASoilV2's more complex soil encoder requires significantly more gradient updates to learn its optimal soil attention weights. This extended training duration is consistent with the more complex optimization landscape introduced by the two-layer soil encoder and learnable attention temperature.

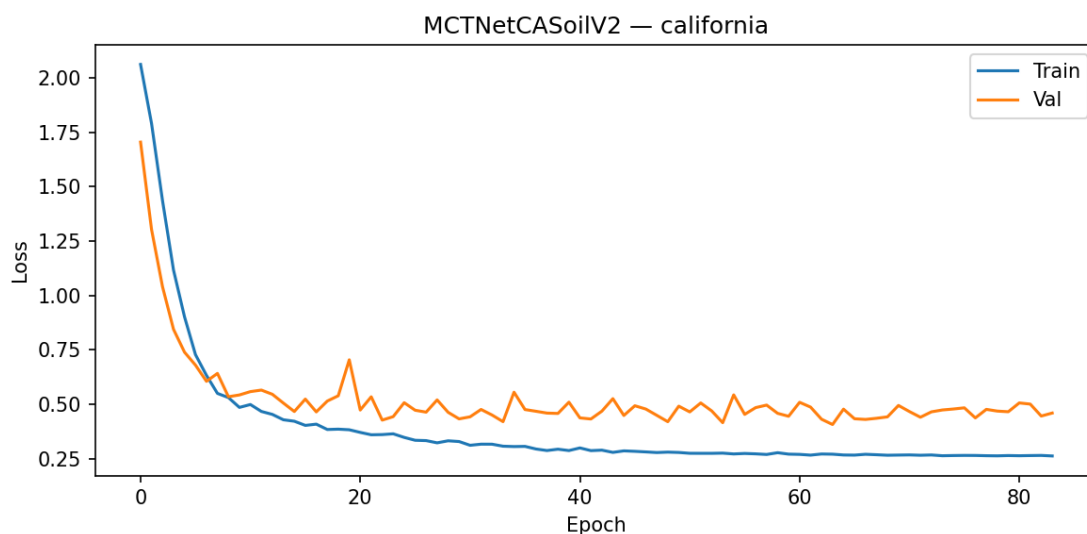


Figure 9-MCTNet-CASoilV2 Training Curve — California.

The V2 California curve is the most challenging training dynamic across all Phase 3 experiments. The validation loss oscillates substantially in the range 0.40–0.70 throughout training, with a pronounced spike to 0.72 around epoch 19 followed by a gradual return to ~0.45 by epoch 85. The training loss descends smoothly to approximately 0.25 by epoch 85. This persistent oscillation in validation performance for California reflects the intrinsic difficulty of the 6-class cross-modal attention learning problem with very few training samples per class. The peaks in validation loss likely correspond to the CosineAnnealingLR re-starts, where the temporarily increased learning rate causes the cross-attention weights to move away from good local minima before the decaying LR phase recovers them. Despite the oscillation, the model identifies a stable good checkpoint around epoch 14–16 (validation loss ~0.45) that is retained by the best-model saving mechanism.

## **B.2 Phase 2 — Environmental Covariate Integration and Ablation Study :**

### **Analysis: Climate Helps Arkansas, Neutral for California:**

MCTNet+Clim improves Arkansas OA by +2.0% but provides zero gain for California ( $\Delta\text{OA} = 0.000$ ,  $\Delta\text{F1} = +0.0006$ , effectively noise). This regional asymmetry is scientifically meaningful and follows directly from the crop compositions of the two regions.

Arkansas features five annual crops: soybeans, rice, corn, cotton, and others. All four named classes are temperature-sensitive annual species whose growing cycles are governed by phenological triggers expressed in degree-day accumulation and precipitation patterns. Corn planted in mid-April requires ~800 GDD above 10°C base to reach silking (July in Arkansas); rice requires sustained temperatures above 20°C from transplanting through heading; cotton is damaged by temperatures below 15°C in the boll-filling period. ERA5-LAND at ~9 km resolution captures the regional climate patterns, the onset of spring warming, summer heat accumulation, and autumn cooling, that differentially phase these phenological triggers. A pixel that accumulates ~800 GDD by late July but not 1,200 GDD is consistent with corn, not rice; this information is encoded in the T2m time series and provides discriminating power beyond what the spectral sequence captures.

California's 6-class problem includes three perennial crops (grapes, almonds, pistachios) whose spectral trajectories are determined by multi-year root and canopy development rather than year-to-year climate fluctuations. A mature almond orchard's NDVI trajectory in 2021 is nearly identical to its trajectory in 2019 and 2023 regardless of whether the year is warm or cool, wet or dry, because the deep root system buffers against short-term climate variability and irrigation compensates for precipitation deficits. Climate variables therefore provide no phenological discrimination among perennial crops. Additionally, California's three annual classes (rice, alfalfa, others) are separated from each other and from the perennials largely by their geometric and structural spectral properties (alfalfa's multi-cut growth pattern visible in the temporal profile, rice's flooded-field appearance in SWIR) rather than by temperature sensitivity. ERA5's ~9 km resolution also flattens what little regional climate variation exists in the spatially compact Sacramento Valley test region.

## B.3 Phase 3 — Proposed Model: MCTNet-CASoil

### Phase 3 Results — CASoil v1:

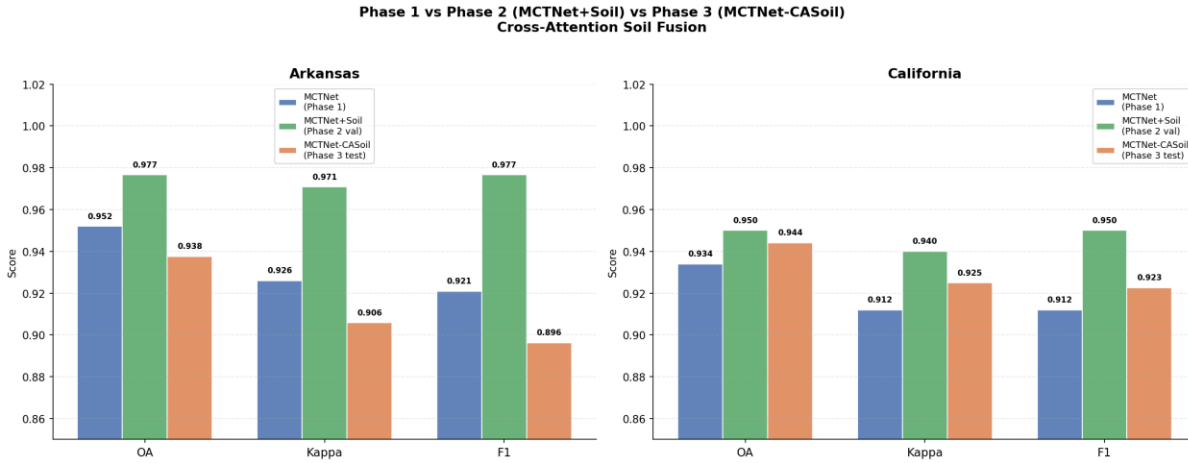


Figure 10: Phase 1 vs Phase 2 vs Phase 3 (CASoil v1) — Cross-Regional Comparison.

The bar chart comparing MCTNet (Phase 1), MCTNet+Soil validation (Phase 2), and MCTNet-CASoil test (Phase 3 v1) across both regions reveals a nuanced result that requires careful interpretation. For Arkansas, CASoil v1 achieves OA=0.938, Kappa=0.906, F1=0.896 on the test set. This is below Phase 1's test OA of 0.952 (−1.4%) and substantially below Phase 2's validation OA of 0.977 (+2.3% vs base). For California, CASoil v1 achieves OA=0.944, Kappa=0.925, F1=0.923, slightly above Phase 1's 0.934 (+1.0%) but below Phase 2 validation's 0.950.

## Appendix C — Extended Experimental Analysis

### C.1 Why Some Crops Are Harder to Classify:

The The per-class F1 results reveal two structural sources of classification difficulty that persist across all models and both phases.

#### Spectral ambiguity between broadleaf annuals:

Soybeans and corn both produce high-NDVI summer canopies with similar red-edge reflectance trajectories. Their primary spectral separation occurs at the temporal endpoints — soybeans are planted 2–3 weeks later than corn in Arkansas (late April vs early April) and harvested 2–3 weeks earlier — creating subtle differences in the temporal position of NDVI emergence and senescence that the CNN branch is designed to capture via its local temporal kernel. However, planting date variability within each class (driven by farmer decision-making, field drainage, and equipment availability) blurs these temporal separators, resulting in the moderate soybean F1 of 0.952 compared to rice's 0.985.

### **The flooded rice advantage:**

Rice achieves the highest per-class F1 across all experiments and both regions (0.985 in Arkansas, 0.997 in California). This is no accident. Rice cultivation in both study regions involves flooding paddies to a depth of 10–20 cm during the transplanting period (May–June in Arkansas, April–May in California). During flooding, SWIR bands (B11, B12) exhibit dramatically low reflectance due to water absorption, creating a spectral signature that is completely unlike any other crop or land cover at that time of year. This SWIR depression is detectable by the CNN branch's local kernels as a sharp temporal transition, providing an unambiguous phenological marker that no other class produces. The MCTNet architecture is, in effect, ideally suited for rice: its CNN branch captures the sharp SWIR transition at flooding onset, and its Transformer branch captures the subsequent NDVI rise as rice plants establish above the water surface.

### **The Others class penalty:**

The consistently low F1 for the Others class (0.839 in Arkansas, 0.939 in California for CASoilV2) reflects a fundamental categorical challenge rather than a model limitation. The Others class includes all pixels not classified as named target crops: fallow land, developed area, water bodies, wetlands, small woodlots, and residual unclassified agricultural pixels. Its spectral distribution is consequently multimodal and incoherent. The model is asked to learn a single class boundary for a category that spans the full spectral space, which is an ill-posed classification problem. High Others recall is achievable by classifying all ambiguous pixels as Others, but this damages precision; high Others precision requires a tight decision boundary that misses atypical Others pixels.

## **C.2 Why Performance Differs Between Arkansas and California:**

Beyond the specific explanations above, several structural factors drive the consistent difference in model performance between the two regions:

### **Training data diversity:**

California's Sacramento Valley is a more geographically compact region with lower within-class spectral variability. The 240 training samples per class more thoroughly represent the spectral distribution of that class's test pixels when the geographic domain is small and uniform. Arkansas, being a sub-region of a highly diverse state, has greater within-class variability that 240 training samples cannot fully capture.

### **Cloud cover impact on critical phenological windows:**

Arkansas's higher missing rate (29.8% vs 24.5%) is particularly damaging for rice classification, July and August, when rice paddies are flooded and the SWIR depression is most diagnostic, are also the peak of the Gulf of Mexico moisture system that drives cloud cover over Arkansas. If the SWIR flooding signature is masked by clouds in key compositing windows, the model loses its most reliable discriminator. The ALPE module mitigates this by suppressing attention to missing positions, but it cannot recover information that was never observed.

### **Perennial vs annual crop dynamics:**

Perennial crops (California) produce temporally smooth, multi-year consistent spectral trajectories that generalize well from the training year to the test set (same year, different pixels). Annual crops (Arkansas) are subject to year-specific weather variations that can shift phenological timing by 2–3 weeks, creating mismatch between training and test spectral patterns even within the same calendar year.

### C.3 Study Limitations:

Several limitations affect the scope and generalizability of the results:

**Evaluation set size:** The 60-sample validation set used in Phase 2 ablation provides limited statistical power. A difference of 2% OA on 360 samples ( $60 \times 6$  classes in California) corresponds to approximately 7 pixels changing classification — within the noise range of any real deep learning experiment. The Phase 2 ablation results should therefore be interpreted as indicative trends rather than statistically robust conclusions.

**Geographic coverage:** Our preprocessing uses 4-tile (Arkansas) and 3-tile (California) Sentinel-2 mosaics covering sub-regions of each state. The paper's results were computed on the full state extent. This limits direct comparison and means our models may not generalize to the full state.

**Cross-attention complexity vs data:** The CASoil cross-attention mechanism introduces additional learnable parameters (the attention projection matrices) that must be learned from 240 samples per class. This is an extremely data-scarce regime for attention learning, and the overfitting observed in the training curves for CASoilV2 (particularly in California) suggests that the model would benefit substantially from a larger labeled training set or from self-supervised pre-training on unlabeled Sentinel-2 data.

**Resolution mismatch for covariates:** As established in Section 4.5, ETOPO1 elevation and ERA5 climate variables are fundamentally mismatched with Sentinel-2's 10 m resolution. The conclusions about their utility would likely change substantially with higher-resolution alternatives.